

Lab3: Dictionary

COCS-221: Computer Science B



Objective

By the end of this lab, you will be able to

- Understand the basic structure and operations of dictionaries in Python.
- Learn how to create, modify, and retrieve data from dictionaries.
- Practice adding, deleting, and updating key-value pairs.
- Explore common dictionary methods like `keys()`, `values()`, `items()`, `get()`, `pop()`, and `clear()`.
- Solve real-world problems using dictionaries for efficient data storage and access.

Exercise

Exercise 1: Basic Dictionary Operations

1. Create a dictionary with 5 student IDs as keys and student names as values.
2. Add two new students to the dictionary.
3. Modify the name of an existing student.
4. Retrieve the value for a student using their ID.
5. Delete a student from the dictionary using their student ID.
6. Try retrieving a non-existing student and demonstrate the `KeyError`.

Exercise 2: Dictionary Methods

1. Use the `keys()` method to display all student IDs.
2. Use the `values()` method to display all student names.

3. Use the `items()` method to display student ID and name pairs.
4. Use the `get()` method to retrieve a student's name and handle a non-existing student.
5. Use the `pop()` method to remove a student by their ID.
6. Use the `clear()` method to remove all entries from the dictionary.

Exercise 3: Real-World Application

1. Create an **empty** contact book simulation using a dictionary where the key is the contact's name and the value is the phone number.
2. **Add 5 contacts** to the dictionary.
3. Implement a **search function** to retrieve a contact's phone number by name.
4. Allow the user to **update** a contact's phone number.
5. Allow the user to **delete** a contact by name.

Exercise 4: Real-World Application

1. Create a gradebook with the following:

"John": 85,

"Mary": 92,

"Paul": 78,

"Anna": 89,

"Steve": 95

2. Create function to calculate the average grade
3. Create function to find the student with the highest grade

Exercise 5: Real-World Application

1. Create a program that takes a sentence as input and counts the frequency of each word in the sentence.
2. Use a dictionary to store the word as the key and the frequency as the value.

Challenging Exercise

1. Problem Description:

You are tasked with creating a system to manage student information for a course. The system will handle:

1. Adding students and their grades.
2. Calculating the average grade of the class.
3. Displaying the student with the highest grade.
4. Displaying the list of students who passed or failed based on their grades.

2. Requirements:

1. Create a dictionary called `courses`, where each key represents a **course name** (e.g., "Math", "Physics", "Chemistry") and the value is another dictionary that stores **student names as keys** and their **grades as values**.
2. Each course should have **3 students**, and the grades should be random values between **0 and 100**.
3. Implement a function of **`average_grade`** that calculates and returns the average grade of all students in a course.

4. Implement a function of `highest_grade` that returns the name of the student with the highest grade in a course.
5. Implement a `passing_students` function that returns a set of names of students who `passed (grades >= 50) in a given course`.
6. Implement a `function failed_students` that returns a set of names of students who failed (grades < 50) in a given course.
7. Verify your result should be:

```
Average grade in Math: 71.0  
Tena has the highest grade in Physics.  
Passing students in Chemistry: {'Dara', 'Monica', 'Pitu', 'Tena', 'Mina'}  
Failed students in Math: {'Pitu'}
```

End

Great job! Keep practicing, and it will get easier. Stay motivated, you're doing great!